A European repository to share neuroimaging datasets publicly.

Dr Cyril Pernet, PhD
Neurobiology Research Unit, Rigshospitalet, DK

publicneuro@nru.dk
cyril.pernet@nru.dk

Septembre 9-10 2026

Public nEUro

INDoS
IMPROVING NEUROIMAGING DATA FOR SHARING

novo nordisk fonden
Benefiting people and society

We are a Data Repository

PublicnEUro.eu is not a research environment (ie it does not compute), it provides an option for researchers to share their data*. (for brain imaging data, only when obtained for research purposes).

Host and publishes (DOI) BIDS datasets, making them findable and accessible.

Public Data: making a dataset information public (open metadata, controlled access for download).

Options for fully open (e.g. animal) or fully closed (embargo period for consortia)

We are a Data Processor

(yes there is a small fee for service and hosting)

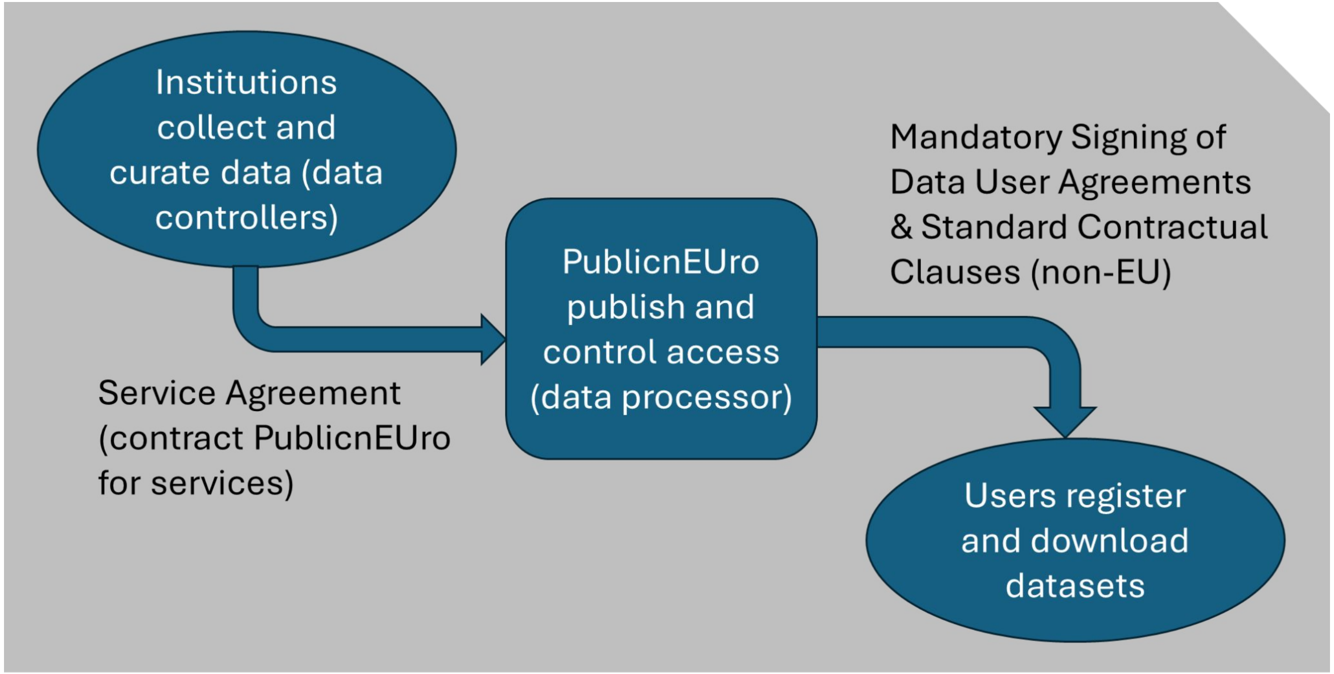


Figure 1: overview of the data workflow (boxes and arrows) and legal tools around it

FAIR data sharing

Finding data within and across repositories

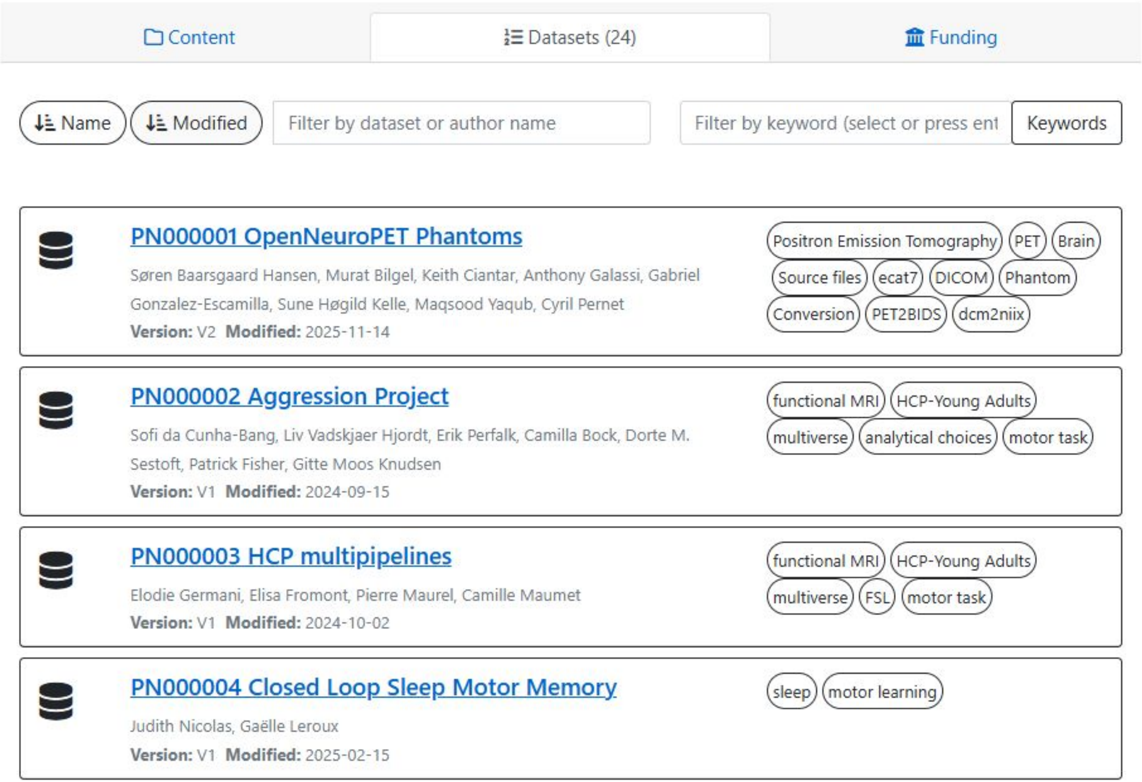
Repository findability: Metadata and metadata curation

Access data in practice: Upload and download

Accessibility policies

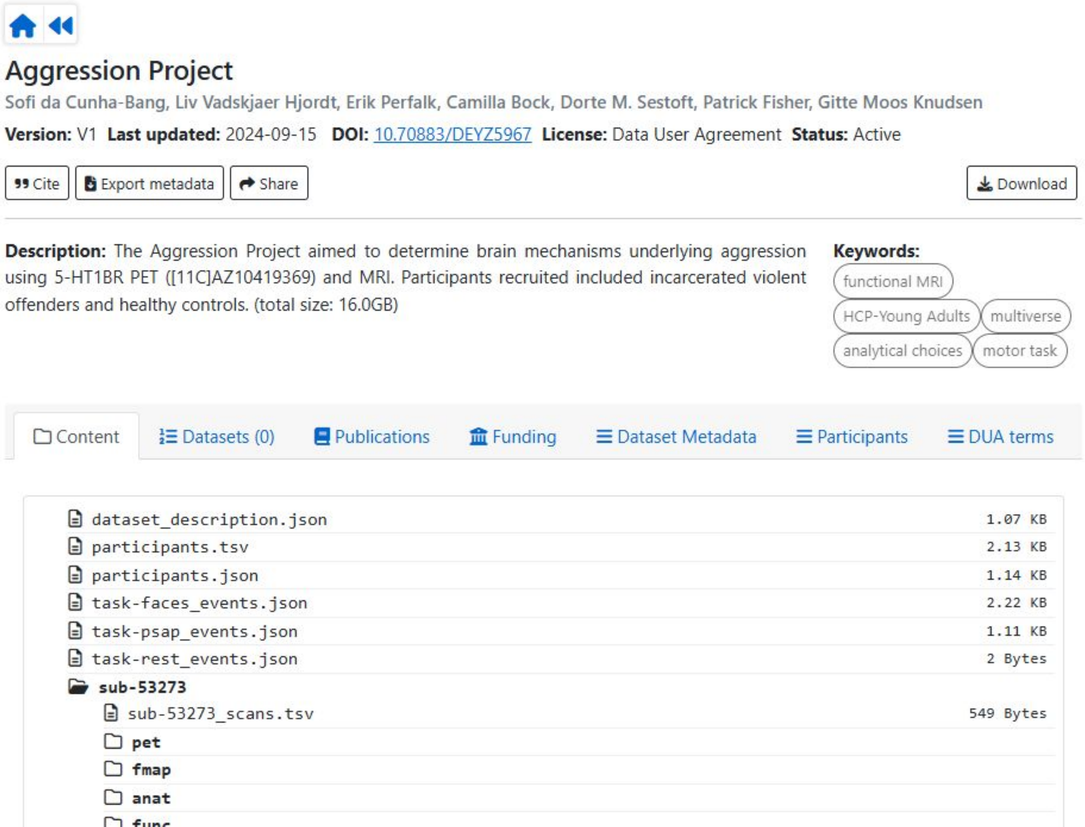
Finding data within and across repositories

Finding data within the repository



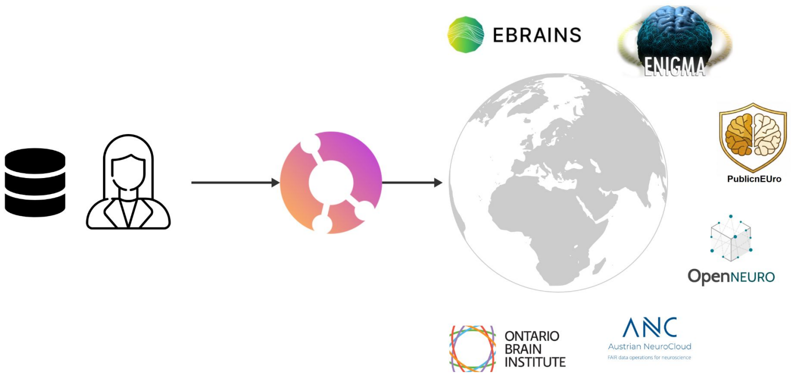
- Browse datasets
- Search by authors
- Search by keywords

Finding data within the repository



- Browsing file representations
- Metadata and Participant summary information

Finding data across repositories



Neurobagel node

-- need to push datasets

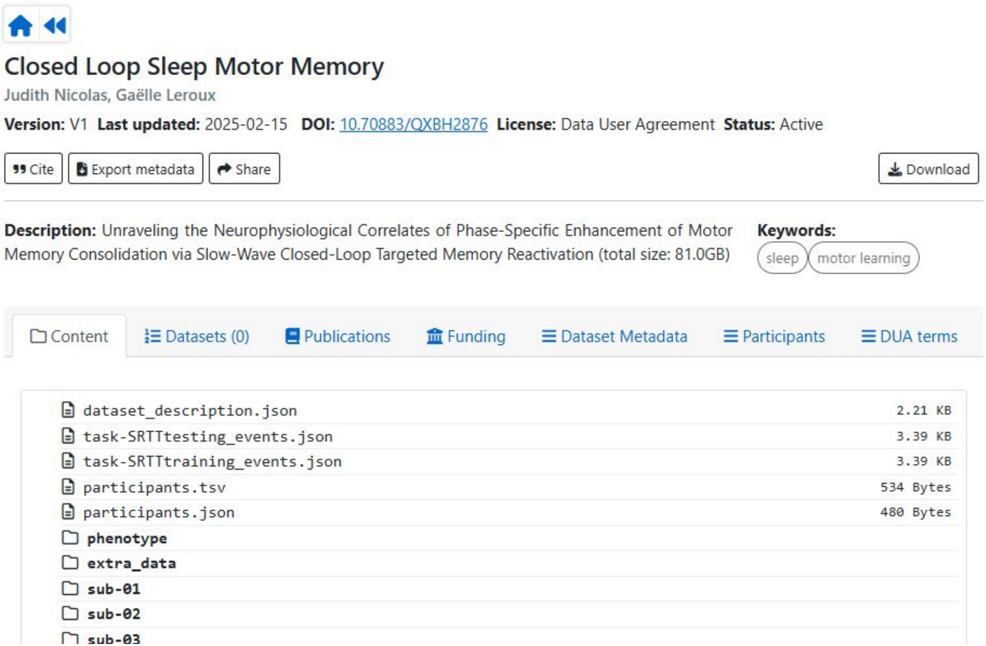
BIDS2EBRAINS

-- need to push datasets and fix issues if any


```
{
  "@context": {"@vocab":
"https://openminds.ebrains.eu/vocab/"},
  "@id":
"https://kg.ebrains.eu/api/instances/<uuid>",
  "@type":
"https://openminds.ebrains.eu/core/FileRepository"
,
  "IRI": "file://local-placeholder",
  "hostedBy": {"@id":
"https://kg.ebrains.eu/instances/organizationalUnit/
ebrains"},
  "name": "local-placeholder"
}
```

Repository findability: Metadata and metadata curation

Concretely



- Each dataset has a description
- PublicnEUro schema also have a **status**
- We provide a DOI
- Datasets come with licenses or DUA → PublicnEUro translates those in D**igital Use Conditions**

https://github.com/Public-nEUro/PublicnEUro-metadata

Closed Loop Sleep Motor Memory

Judith Nicolas, Gaëlle Leroux

Version: V1 **Last updated:** *unknown* **DOI:** 10.70883/QXBH2876 **License:** Data User Agreement

Cite | Export metadata | Share | Download

Description: Unraveling the Neurophysiological Correlates of Phase-Specific Enhancement of Motor Memory Consolidation via Slow-Wave Closed-Loop Targeted Memory Reactivation

Keywords:
sleep | motor learning

Content | Datasets (0) | Publications | Funding | Dataset Metadata | Participants | DUA terms

Property	Value(s)
Bids_version	v1.9.0
Bids_datasettype	raw
Bids_datatypes	anat func eeg
NCBI Species Taxonomy	homo sapiens
Cognitive Atlas concept	motor sequence learning
Cognitive Atlas task	serial reaction time task rest eyes open Pittsburgh Sleep Quality Index

Closed Loop Sleep Motor Memory

Judith Nicolas, Gaëlle Leroux

Version: V1 **Last updated:** *unknown* **DOI:** 10.70883/QXBH2876 **License:** Data User Agreement

Cite | Export metadata | Share | Download

Description: Unraveling the Neurophysiological Correlates of Phase-Specific Enhancement of Motor Memory Consolidation via Slow-Wave Closed-Loop Targeted Memory Reactivation

Keywords: sleep, motor learning

Content | Datasets (0) | Publications | Funding | Dataset Metadata | Participants | DUA terms

Property	Value(s)
Total_number	31
Age_range	[18, 30]
Number_of_healthy	31
Number_of_biological_males	16
Number_of_biological_females	15

PublicnEUro metadata

Machine-readable, dataset-level governance metadata derived from the PublicnEUro DataCatalogue.

The generated records keep three independent facts for every dataset version:

- **status:** `active`, `archived`, `retired`, `withdrawn`, or `superseded`;
- **retrieval mode:** `online`, `cold_archive`, `external`, or `unavailable`;
- **digital use conditions (DUC):** the licence or Data User Agreement summary exposed by the catalogue.

For backwards compatibility, a catalogue record without `status` is exported as `active`.

24 datasets / 25 versions

- **Access:** 3 open access / 21 restricted
- **Participants:** 2,130 total (1,731 healthy / 399 patients)
- **Documented size:** 5.06 TB (reported for 23/24 datasets)

Figures use the latest version of each dataset. Access is restricted when the catalogue licence is `Data User Agreement`; participant counts come from the `Participants` panel; patients are inferred as total minus healthy participants. Counts are dataset-level and may include the same individuals in related datasets.

Dataset	Version	Status	Retrieval	DUC	DOI	Updated
PN000001 OpenNeuroPET Phantoms	V1	superseded	online	1 reviewed condition	10.70883/IGQP1334	2024-06-02
PN000001 OpenNeuroPET Phantoms	V2	active	online	1 reviewed condition	10.70883/RGNV3768	2025-11-14
PN000002 Aggression Project	V1	active	online	9 reviewed conditions	10.70883/DEYZ5967	2024-09-15
PN000003 HCP multipipelines	V1	active	online	7 reviewed conditions	10.70883/GTKK1541	2024-10-02
PN000004 Closed Loop Sleep Motor Memory	V1	active	online	6 reviewed conditions	10.70883/QXBH2876	2025-02-15

- curation
- datasets
  - PN000001.json
  - PN000002.json
  - PN000003.json
  - PN000004.json
  - PN000005.json
  - PN000006.json
  - PN000007.json
  - PN000008.json
  - PN000009.json
  - PN000010.json

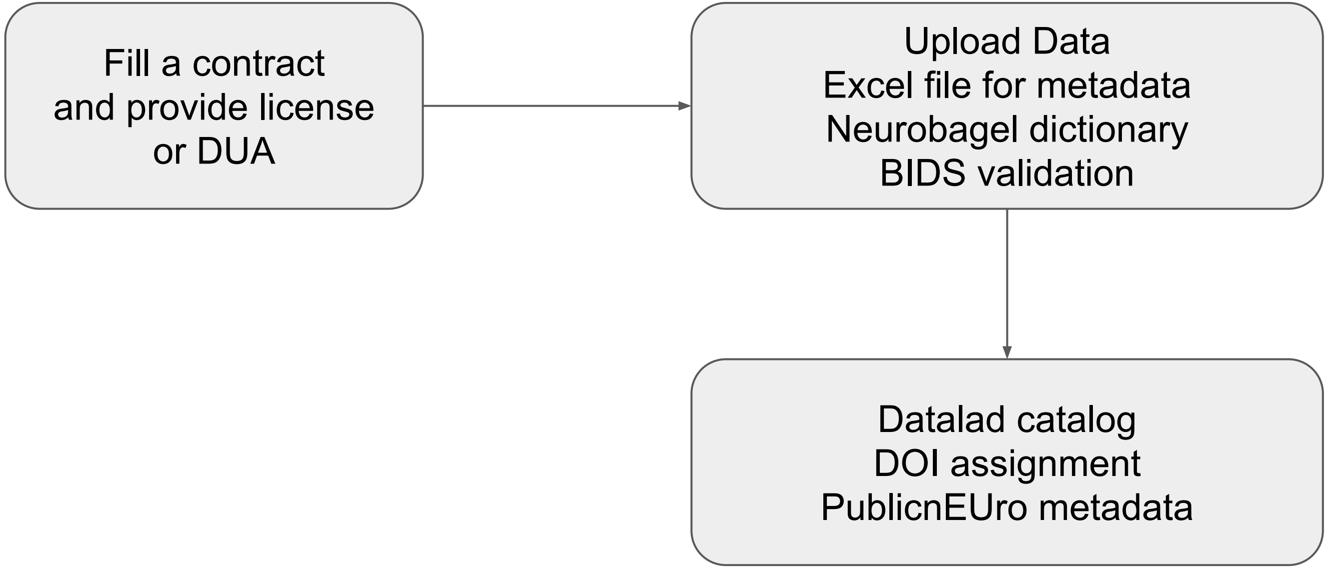

```
{
  "schemaVersion": "1.3",
  "datasetId": "PN000002",
  "name": "Aggression Project",
  "versions": [
    {
      "version": "V1",
      "catalogueUrl": "https://datacatalog.publicneuro.eu/dataset/PI
      "status": "active",
      "retrieval": {
        "mode": "online",
        "url": "https://datacatalog.publicneuro.eu/manage/request-a
      },
      "duc": {
        "profileVersion": "0.1.0",
        "profileName": "PN000002 V1 Digital Use Conditions",
        "ducVersion": "1.1.0",
        "permissionMode": "All unstated conditions are Permitted",
        "language": "eng",
        "assets": [
```


- curation
- datasets
- exports
  - openaire-cerif.xml
  - re3data.xml
  - repository-summary.json
- mappings
  - README.md
  - dua_to_duc.json
- schema
- scripts
- tests
- .gitignore

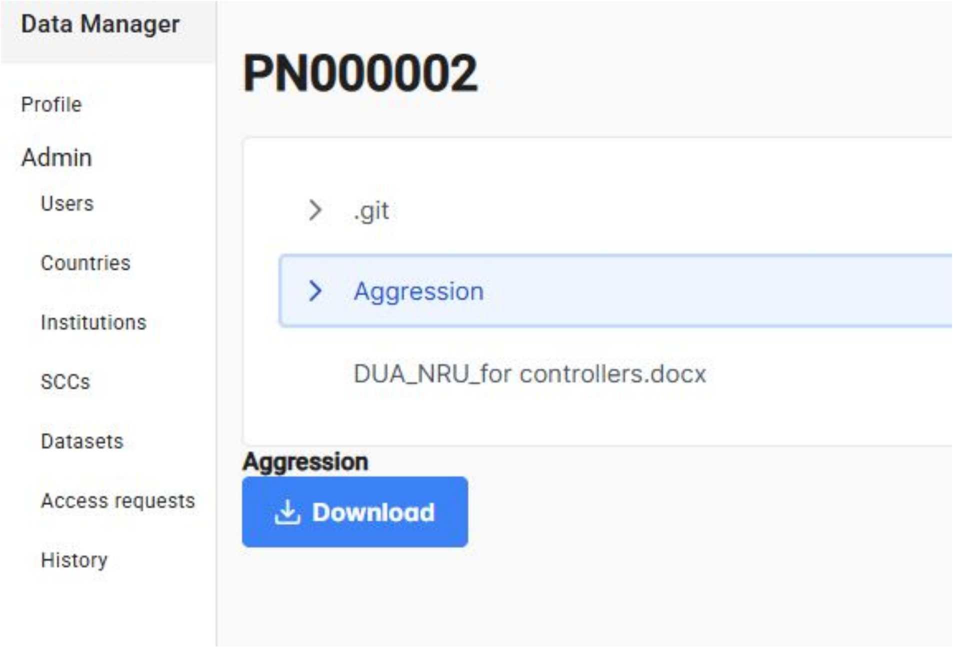

```
{
  "schemaVersion": "1.0",
  "datasets": 24,
  "versions": 25,
  "access": {
    "open": 3,
    "restricted": 21,
    "unclassified": 0
  },
  "participants": {
    "total": 2130,
    "healthy": 1731,
    "patients": 399,
    "datasetsWithCounts": 23
  },
  "documentedSize": {
    "terabytes": 5.06,
    "datasetsWithSize": 23
  }
}
```

Access data in practice: Upload and Download

Upload is done only via GUI / drag and drop



2 options to access data



Or CLI

bash <(wget -qO- https://datacatalog.publicneuro.eu/api/cli.sh)

This will prompt you to enter

your email

your password

the PublicnEUro (PN) ID

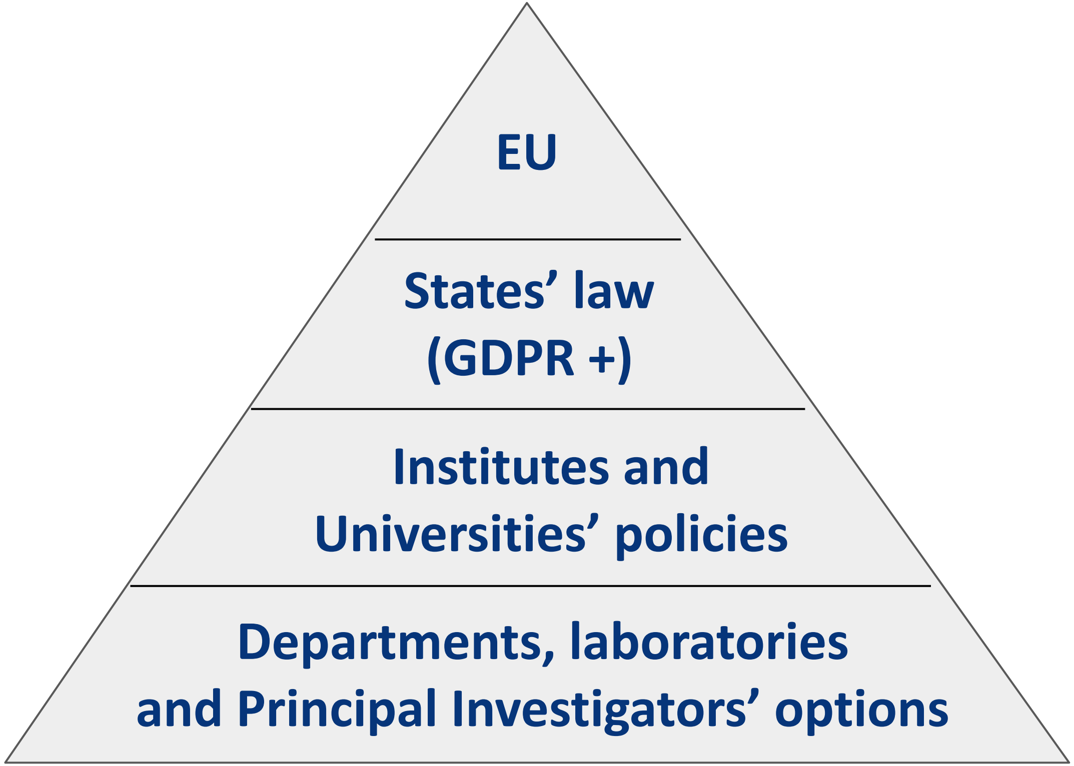
the folder that you want to download - use / for the entire dataset

Accessibility policies

GDPR, states, institutions and labs

GDPR declares that within EU personal data can be shared (Art 1), it also states that general conditions do not apply to research data (Art 5 and recital 50), i.e. scientific data can be shared more broadly without purpose limitation (despite layers reluctance).

YOU CAN SHARE - but each state, institution, lab has their own implementation trickling down to you!



GDPR, states, institutions and labs

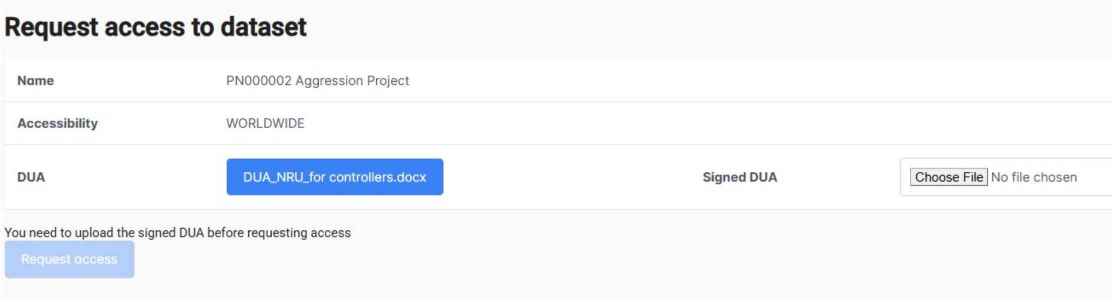
Each state has a specific implementation adding different restrictions. Each institution has specific policies and data Protection Officers interpret the law and those policies differently. Each lab has specific views on how they want to control data.

By controlling the data users geolocation and by letting data controllers set their ‘terms and conditions’, PublicnEUro accommodates the different regulations.

Restrictions	Give data processor rights (e.g. with oversight)	Give data controller rights (e.g. without oversight)
EU / EU+	Most restrictive	
World wide		Most permissive

Doc signing

SSC are institutional and exchanged by email and stored - Institutions are then marked as accepting SSC



DUA can be signed online or uploaded (data controller validated)